

Technology Stack

Date	25 June 2026
Project Name	Smart Lender
Maximum Marks	2 Marks

Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that will be used to build the solution. A well-chosen technology stack ensures that the application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, Bootstrap	Used to create a responsive, user-friendly, and visually appealing web interface for entering loan details and displaying prediction results.

2	Backend / Server-Side	Python, Flask	Flask handles HTTP requests, processes user input, interacts with the machine learning model, and returns prediction results efficiently.
3	Database / Data Storage	CSV Dataset & Pickle(.pkl) Files	The CSV dataset was used for training, while the trained XGBoost model and scaler are stored as Pickle files for fast prediction without retraining.
4	Cloud / Hosting / Deployment	IBM Cloud(Cloud Foundry)	Provides reliable cloud deployment, easy accessibility, and allows users to access the application from anywhere through a web browser.
5	Version Control & CI/CD	Git & Github	Git tracks code changes, while GitHub enables collaboration, version management, and project backup.
6	Third-Party APIs / Other Tools	Scikit-learn, XGBoost, Pandas, NumPy, Joblib	Pandas and NumPy handle data processing, Scikit-learn is used for preprocessing and evaluation, XGBoost performs loan prediction, and Joblib stores the trained model and scaler efficiently.